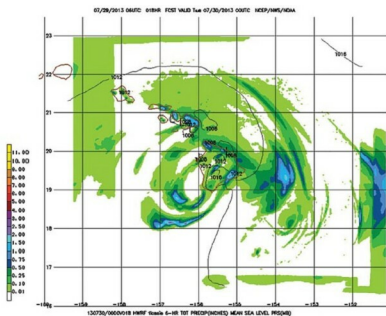


Role of IT in WEATHER FORECASTING



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Fig. 1: Snapshot of Hurricane Research and Forecasting (HRRF), one of the most sophisticated numerical computer models now being run on NOAA's supercomputers (Source: www.noaa.gov)



The weather has a profound impact on a person's mood. Research shows that higher temperatures raise a person's mood, while weather encompassing wind or little sun tends to make a person feel depressed. As temperature rises, intergroup conflicts and interpersonal violence are known to increase, even though only marginally. These findings hold well not only for higher temperatures but also for rainfall.

Global warming is being caused by an increase in the average temperature of Earth's surface. For life to exist on Earth, an average temperature of 15°C is maintained by the natural greenhouse effect. Without this effect, the average temperature would be -18°C. This average temperature rose by about 0.9°C during the last century. Scientific studies indicate that average temperature will rise further by 1.1°C to 4.5°C during the 21st century, depending upon an increase in greenhouse gases emissions. As an result, net cereal production in South Asian countries is projected to decline by four per cent to ten percent by the end of this century.

Melting glaciers could seriously affect five billion people in Himalaya-Hindu-Kush region. There could be more intense rainfall resulting in floods and more dry days in a year, causing drought. Deaths due to heat waves, water-borne and vector-borne diseases like dengue are also expected to increase. The tsunami that occurred in 2004 took millions of lives and caused destruction worth billions of dollars.

This is merely a glimpse into the repercussions of climatic changes and it makes us realise how important it is to be able to predict the weather. Powerful computers are playing a key role in making accurate weather and meteorological forecasts. However, Earth's atmosphere follows complex non-linear rules of fluid dynamics. Despite meteorologists having comprehensive knowledge of the current weather, it is actually challenging to calculate a forecast before the next change in the weather actually happens.

Therefore there is need for more sophisticated information technology (IT) in the form of supercomputers and advanced software to handle complex weather equations. For example, National Oceanic and Atmospheric Association (NOAA) uses a supercomputer called Theia, which can make as many as quadrillion calculations per second and has 28,000 computer cores.

Computer cores enable very-high-speed calculations. Each computer chip has multiple cores, so the chip can do multiple calculations simultaneously. In a supercomputer, the chips are interconnected. Hence a task like predicting the weather of New Delhi a week from now is split among thousands of chips and tens of thousands of computer cores.

Historical role played by IT

Back in 1916, World War I ambulance driver, Lewis Fry Richardson, imagined a worldwide system of weather prediction in which Earth was divided into 64,000 cells. Each cell was to report its weather